

1. Create user with name Techie and provide sudo access

Command Executed:

```
sudo useradd khadeer
```

```
sudo passwd khadeer@
```

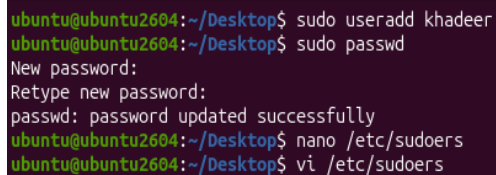
```
sudo su khadeer
```

```
vi /etc/sudoers
```

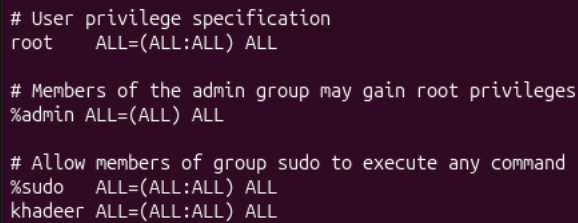
Command Output:

User created successfully, password set, and user added to sudo group.

Evidence (Screenshot):



```
ubuntu@ubuntu2604:~/Desktop$ sudo useradd khadeer
ubuntu@ubuntu2604:~/Desktop$ sudo passwd
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ubuntu2604:~/Desktop$ nano /etc/sudoers
ubuntu@ubuntu2604:~/Desktop$ vi /etc/sudoers
```



```
# User privilege specification
root    ALL=(ALL:ALL) ALL

# Members of the admin group may gain root privileges
%admin   ALL=(ALL) ALL

# Allow members of group sudo to execute any command
%sudo   ALL=(ALL:ALL) ALL
khadeer  ALL=(ALL:ALL) ALL
```

Key Learning:

- useradd creates a new user, passwd sets the password.
- and sudo su changes the users

2. Navigate to the home directory

The terminal shows the home directory path.

Command Executed:

```
cd ~
```

```
pwd
```

Command Output:

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~/Desktop$ pwd
/home/ubuntu/Desktop
ubuntu@ubuntu2604:~/Desktop$ cd ~
ubuntu@ubuntu2604:~$ pwd
/home/ubuntu
```

Key Learning:

cd ~ takes you directly to the home directory.

3. Create a new directory

Command Executed:

mkdir khadeer

ls

Command Output:

A new directory named khadeer is created.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  snap
ubuntu@ubuntu2604:~$ mkdir khadeer
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  snap
```

Key Learning:

- mkdir is used to create directories.
 - ls is used to list the directories
-

4. List the contents of a directory

Command Executed:

ls -l

Command Output:

Files and folders are displayed in long format.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls -l
total 40
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 26 10:23 khadeer
drwx----- 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
```

Key Learning:

ls -l shows permissions, owner, size, and time.

5. Change the current directory

Command Executed:

```
cd /home/ubuntu/Desktop
```

```
pwd
```

Command Output:

Current directory changes to Desktop.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ pwd
/home/ubuntu
ubuntu@ubuntu2604:~$ cd /home/ubuntu/Desktop
ubuntu@ubuntu2604:~/Desktop$ pwd
/home/ubuntu/Desktop
ubuntu@ubuntu2604:~/Desktop$
```

Key Learning:

cd changes the working directory.

6. Create a new empty file

Command Executed:

```
touch khadeer.txt
```

```
ls -l
```

Command Output:

An empty file named khadeer.txt is created.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ touch khadeer.txt
ubuntu@ubuntu2604:~$ ls -l
total 40
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
drwxrwxr-x 2 ubuntu ubuntu 4096 Jun 26 10:23 khadeer
-rw-rw-r-- 1 ubuntu ubuntu 0 Jun 26 10:53 khadeer.txt
drwx----- 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
```

Key Learning:

touch creates empty files.

7. View the contents of a file

Command Executed:

```
cat khadeer.txt
```

```
ls
```

Command Output:

```
Hi this khadeer
```

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  khadeer.txt  snap
ubuntu@ubuntu2604:~$ cat khadeer.txt
Hi this khadeer
```

Key Learning:

cat is used to view small text files.

8. Copy a file to another location

Command Executed:

```
cp khadeer.txt /home/ubuntu/Desktop
```

```
ls
```

Command Output:

The file is copied to the parent directory to Desktop directory.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos khadeer khadeer.txt snap
ubuntu@ubuntu2604:~$ cp khadeer.txt /Documents
cp: cannot create regular file '/Documents': Permission denied
ubuntu@ubuntu2604:~$ cp khadeer.txt /home/ubuntu/Desktop
ubuntu@ubuntu2604:~$ cd /home/ubuntu/Desktop
ubuntu@ubuntu2604:~/Desktop$ pwd
/home/ubuntu/Desktop
ubuntu@ubuntu2604:~/Desktop$ ls
khadeer.txt
```

Key Learning:

cp copies files.

9. Move a file to another location

Command Executed:

```
mv file1.txt ../
```

```
ls
```

```
cd Desktop
```

Command Output:

The file is moved to another location.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos khadeer snap
ubuntu@ubuntu2604:~$ mv khadeer /home/ubuntu/Desktop
ubuntu@ubuntu2604:~$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos snap
ubuntu@ubuntu2604:~$ cd Desktop
ubuntu@ubuntu2604:~/Desktop$ ls
khadeer
```

10. Rename a file

Command Executed:

```
mv khadeer.txt shaik.txt
```

```
ls
```

Command Output:

The file is renamed khadeer.txt to shaik.txt

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  khadeer.txt  snap
ubuntu@ubuntu2604:~$ mv khadeer.txt shaik.txt
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  shaik.txt  snap
```

Key Learning:

- mv is also used to rename files.
-

11. Delete a file

Command Executed:

```
rm khadeer.txt
```

```
ls
```

Command Output:

The file is deleted Successfully.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  shaik.txt  snap
ubuntu@ubuntu2604:~$ rm shaik.txt
ubuntu@ubuntu2604:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  khadeer  snap
ubuntu@ubuntu2604:~$
```

Key Learning:

- rm command used to delete the Files.
 - rm deletes files permanently.
-

12. Grant or revoke permissions on a file or directory

Command Executed:

```
chmod 777 khadeer
```

```
ls -l
```

Command Output:

Permissions are changed.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ls -l
total 40
drwxr-xr-x 2 ubuntu ubuntu 4096 Jun 27 07:17 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
drw-rw-r-- 2 ubuntu ubuntu 4096 Jun 27 07:29 khadeer
drwx----- 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
ubuntu@ubuntu2604:~$ chmod 777 khadeer
ubuntu@ubuntu2604:~$ ls -l
total 40
drwxr-xr-x 2 ubuntu ubuntu 4096 Jun 27 07:17 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
drwxrwxrwx 2 ubuntu ubuntu 4096 Jun 27 07:29 khadeer
drwx----- 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
ubuntu@ubuntu2604:~$
```

Key Learning:

- chmod changes read, write, and execute permissions.
- chmod may be useful in some cases of permission changes.

13. View the current date and time

Command Executed:

date

Command Output:

Current date and time are displayed.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ date
Sat Jun 27 07:42:46 AM EDT 2026
ubuntu@ubuntu2604:~$
```

Key Learning:

- date shows system date and time.

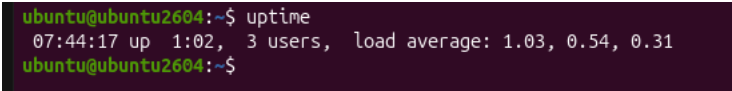
14. Check the system uptime

Command Executed:

uptime

Command Output:

System uptime and load average are shown.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ uptime
07:44:17 up 1:02, 3 users, load average: 1.03, 0.54, 0.31
ubuntu@ubuntu2604:~$
```

Key Learning:

- uptime shows how long the system has been running.
-

15. View the running processes**Command Executed:**

ps -ef

Command Output:

Running processes are listed.

Evidence (Screenshot):

```
07:47:17 up 17:02, 5 users, load average: 1.03, 0.14, 0.01
ubuntu@ubuntu2604:~$ ps -ef
UID          PID    PPID  C   STIME TTY          TIME CMD
root           1        0  0  06:41 ?        00:00:12 /usr/lib/systemd/systemd --switched-root --system --des
root           2        0  0  06:41 ?        00:00:00 [kthreadd]
root           3        2  0  06:41 ?        00:00:00 [pool_workqueue_release]
root           4        2  0  06:41 ?        00:00:00 [kworker/R-rcu_gp]
root           5        2  0  06:41 ?        00:00:00 [kworker/R-sync_wq]
root           6        2  0  06:41 ?        00:00:00 [kworker/R-kvfree_rcu_reclaim]
root           7        2  0  06:41 ?        00:00:00 [kworker/R-slub_flushwq]
root           8        2  0  06:41 ?        00:00:00 [kworker/R-netns]
root          12        2  0  06:41 ?        00:00:00 [kworker/u8:0-ipv6_addrconf]
root          13        2  0  06:41 ?        00:00:00 [kworker/R-mm_percpu_wq]
root          14        2  0  06:41 ?        00:00:02 [ksoftirqd/0]
root          15        2  0  06:41 ?        00:00:02 [rcu_preempt]
root          16        2  0  06:41 ?        00:00:00 [rcu_exp_par_gp_kthread_worker/0]
root          17        2  0  06:41 ?        00:00:00 [rcu_exp_gp_kthread_worker]
root          18        2  0  06:41 ?        00:00:00 [migration/0]
root          19        2  0  06:41 ?        00:00:00 [kprobe-optimizer]
root          20        2  0  06:41 ?        00:00:00 [idle_inject/0]
root          21        2  0  06:41 ?        00:00:00 [cpuhp/0]
root          22        2  0  06:41 ?        00:00:00 [cpuhp/1]
root          23        2  0  06:41 ?        00:00:00 [idle_inject/1]
root          24        2  0  06:41 ?        00:00:00 [migration/1]
root          25        2  0  06:41 ?        00:00:01 [ksoftirqd/1]
root          26        2  0  06:41 ?        00:00:01 [kworker/1:0-cgroup_free]
root          29        2  0  06:41 ?        00:00:03 [kworker/u10:0-events_unbound]
root          30        2  0  06:41 ?        00:00:00 [kdevtmpfs]
root          31        2  0  06:41 ?        00:00:00 [kworker/R-inet_frag_wq]
root          32        2  0  06:41 ?        00:00:00 [rcu_tasks_kthread]
root          33        2  0  06:41 ?        00:00:00 [rcu_tasks_rude_kthread]
root          34        2  0  06:41 ?        00:00:00 [kauditd]
root          35        2  0  06:41 ?        00:00:00 [khungtaskd]
root          36        2  0  06:41 ?        00:00:00 [oom_reaper]
root          39        2  0  06:41 ?        00:00:00 [kworker/R-writeback]
root          40        2  0  06:41 ?        00:00:00 [kcompactd0]
root          41        2  0  06:41 ?        00:00:00 [ksmd]
root          42        2  0  06:41 ?        00:00:00 [khugepaged]
root          43        2  0  06:41 ?        00:00:00 [kworker/R-kblockd]
```

Key Learning:

- ps -ef displays active processes.
- ps -ef also displays UID,PID,PPID.

16. Kill a running process

Command Executed:

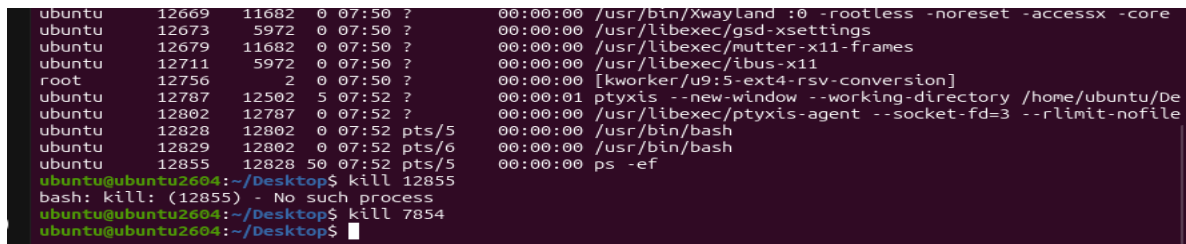
```
ps -ef
```

```
kill 7854
```

Command Output:

The selected process stops.

Evidence (Screenshot):



A terminal window showing a list of running processes. The processes include Xwayland, gsd-xsettings, mutter-x11-frames, ibus-x11, [kworker/u9:5-ext4-rsv-conversion], ptyxis, ptyxis-agent, and bash. The user then enters the command 'kill 12855', which results in a message 'bash: kill: (12855) - No such process'. Finally, the user enters 'kill 7854'.

```
ubuntu 12669 11682 0 07:50 ? 00:00:00 /usr/bin/Xwayland :0 -rootless -noreset -accessx -core
ubuntu 12673 5972 0 07:50 ? 00:00:00 /usr/libexec/gsd-xsettings
ubuntu 12679 11682 0 07:50 ? 00:00:00 /usr/libexec/mutter-x11-frames
ubuntu 12711 5972 0 07:50 ? 00:00:00 /usr/libexec/ibus-x11
root 12756 2 0 07:50 ? 00:00:00 [kworker/u9:5-ext4-rsv-conversion]
ubuntu 12787 12502 5 07:52 ? 00:00:01 ptyxis --new-window --working-directory /home/ubuntu/De
ubuntu 12802 12787 0 07:52 ? 00:00:00 /usr/libexec/ptyxis-agent --socket-fd=3 --rlimit-nofile
ubuntu 12828 12802 0 07:52 pts/5 00:00:00 /usr/bin/bash
ubuntu 12829 12802 0 07:52 pts/6 00:00:00 /usr/bin/bash
ubuntu 12855 12828 50 07:52 pts/5 00:00:00 ps -ef
ubuntu@ubuntu2604:~/Desktop$ kill 12855
bash: kill: (12855) - No such process
ubuntu@ubuntu2604:~/Desktop$ kill 7854
ubuntu@ubuntu2604:~/Desktop$
```

Key Learning:

- kill stops a process using its PID.
-

17. Install a package using the package manager

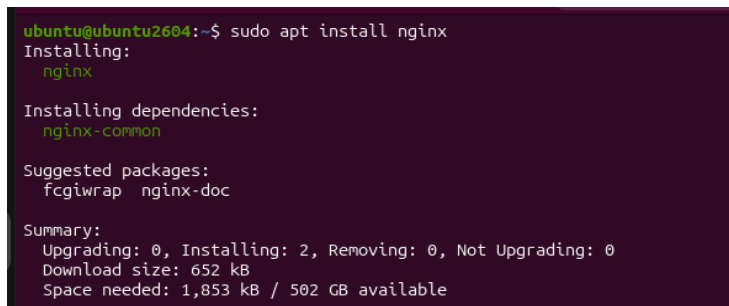
Command Executed:

sudo apt install nginx

Command Output:

Package gets installed.

Evidence (Screenshot):



A terminal window showing the command 'sudo apt install nginx'. The output shows the installation of nginx and its dependencies, including nginx-common. It also lists suggested packages: fcgiwrap and nginx-doc. A summary at the bottom shows the download size and space needed.

```
ubuntu@ubuntu2604:~$ sudo apt install nginx
Installing:
  nginx

Installing dependencies:
  nginx-common

Suggested packages:
  fcgiwrap  nginx-doc

Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 0
  Download size: 652 kB
  Space needed: 1,853 kB / 502 GB available
```

Key Learning:

apt installs packages on Debian-based systems.

18. Update the system packages

Command Executed:

sudo apt update

Command Output:

System package lists are updated.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ sudo apt update
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Ign:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Ign:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Err:3 http://archive.ubuntu.com/ubuntu resolute-backports InRelease
Temporary failure resolving 'archive.ubuntu.com'
Err:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Temporary failure resolving 'security.ubuntu.com'
Err:2 http://archive.ubuntu.com/ubuntu resolute-updates InRelease
Temporary failure resolving 'archive.ubuntu.com'
Err:1 http://archive.ubuntu.com/ubuntu resolute InRelease
Temporary failure resolving 'archive.ubuntu.com'
All packages are up to date.
```

Key Learning:

Updating keeps the system secure and current.

19. Create a symbolic link

A symbolic link, or symlink, is like a shortcut that points to another file or folder instead of containing the actual data itself.

Command Executed:

```
ln -s /home/ubuntu/khadeer.txt khadeer
```

```
ls -l
```

Command Output:

A symbolic link is created.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ ln -s /home/ubuntu/khadeer.tx khadeer
ubuntu@ubuntu2604:~$ ls -l
total 36
drwxr-xr-x 3 ubuntu ubuntu 4096 Jun 29 08:09 Desktop
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Documents
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Downloads
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Music
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Pictures
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Public
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Templates
drwxr-xr-x 2 ubuntu ubuntu 4096 Apr 28 07:14 Videos
lrwxrwxrwx 1 ubuntu ubuntu 23 Jun 29 08:30 khadeer -> /home/ubuntu/khadeer.tx
-rw-rw-r-- 1 ubuntu ubuntu 0 Jun 29 08:19 khadeer.txt
-rw-rw-r-- 1 ubuntu ubuntu 0 Jun 29 08:25 shaik.txt
drwx----- 5 ubuntu ubuntu 4096 Jun 26 03:00 snap
```

Key Learning:

ln -s creates a soft link.

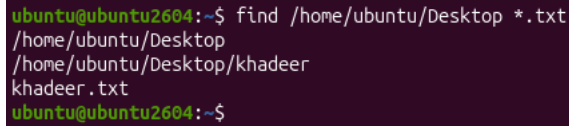
20. Search for files using the find command

Command Executed:

```
find /home/ubuntu/Desktop *.txt
```

Command Output:

Matching files are displayed.

Evidence (Screenshot):

```
ubuntu@ubuntu2604:~$ find /home/ubuntu/Desktop *.txt
/home/ubuntu/Desktop
/home/ubuntu/Desktop/khadeer
khadeer.txt
ubuntu@ubuntu2604:~$
```

Key Learning:

find searches files and folders.

21- Compress and decompress files using tar.

tar is used to bundle files and folders into one archive, and it can also compress or extract that archive in one step.

Command Executed:

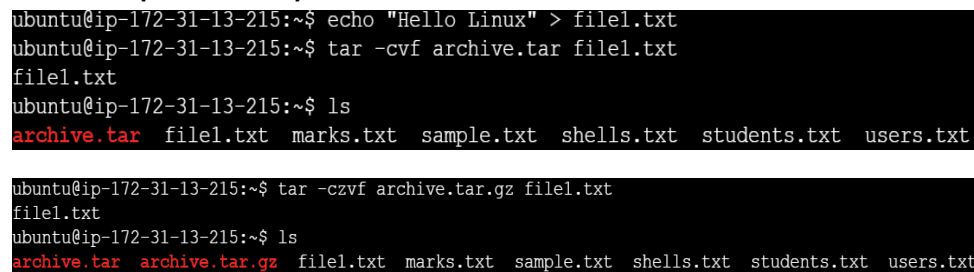
```
echo "Hello Linux" > file1.txt
```

```
-tar -cvf archive.tar file1.txt
```

```
tar -czvf archive.tar.gz file.txt
```

Command Output:

Archive is created and extracted.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ echo "Hello Linux" > file1.txt
ubuntu@ip-172-31-13-215:~$ tar -cvf archive.tar file1.txt
file1.txt
ubuntu@ip-172-31-13-215:~$ ls
archive.tar  file1.txt  marks.txt  sample.txt  shells.txt  students.txt  users.txt

ubuntu@ip-172-31-13-215:~$ tar -czvf archive.tar.gz file1.txt
file1.txt
ubuntu@ip-172-31-13-215:~$ ls
archive.tar  archive.tar.gz  file1.txt  marks.txt  sample.txt  shells.txt  students.txt  users.txt
```

Key Learning:

tar is used for archiving and extraction.

22. Monitor system resources with top or htop

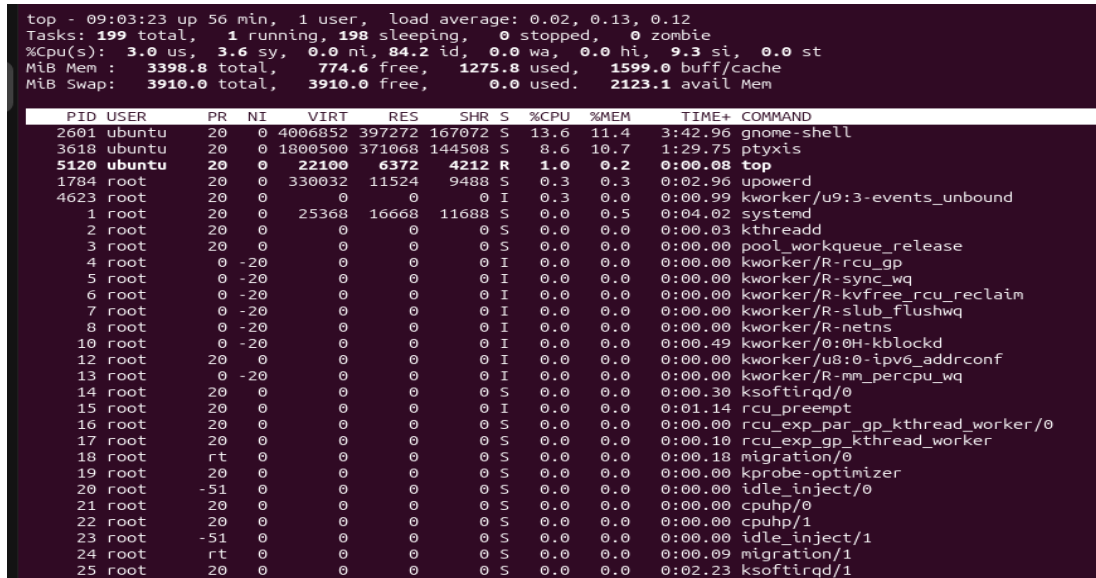
Command Executed:

top

Command Output:

Live CPU, memory, and process usage shown.

Evidence (Screenshot):



```
top - 09:03:23 up 56 min, 1 user, load average: 0.02, 0.13, 0.12
Tasks: 199 total, 1 running, 198 sleeping, 0 stopped, 0 zombie
%cpu(s):  3.0 us,  3.6 sy,  0.0 ni, 84.2 id,  0.0 wa,  0.0 hi,  9.3 si,  0.0 st
MiB Mem : 3398.8 total,  774.6 free, 1275.8 used, 1599.0 buff/cache
MiB Swap: 3910.0 total, 3910.0 free,  0.0 used, 2123.1 avail Mem
```

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
2601	ubuntu	20	0	4006852	397272	167072	S	13.0	11.4	3:42.96	gnome-shell
3618	ubuntu	20	0	1800500	371068	144508	S	8.6	10.7	1:29.75	ptxsis
5120	ubuntu	20	0	22100	6372	4212	R	1.0	0.2	0:00.08	top
1784	root	20	0	330032	11524	9488	S	0.3	0.3	0:02.96	upowerd
4623	root	20	0	0	0	0	I	0.3	0.0	0:00.99	kworker/u9:3-events_unbound
1	root	20	0	25368	16668	11688	S	0.0	0.5	0:04.02	systemd
2	root	20	0	0	0	0	S	0.0	0.0	0:00.03	kthreadd
3	root	20	0	0	0	0	S	0.0	0.0	0:00.00	pool_workqueue_release
4	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-rcu_gp
5	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-sync_wq
6	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-kvfree_rcu_reclaim
7	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-slub_flushwq
8	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-netns
10	root	0	-20	0	0	0	I	0.0	0.0	0:00.49	kworker/0:0H-kblockd
12	root	20	0	0	0	0	I	0.0	0.0	0:00.00	kworker/u8:0-ipv6_addrconf
13	root	0	-20	0	0	0	I	0.0	0.0	0:00.00	kworker/R-mm_percpu_wq
14	root	20	0	0	0	0	S	0.0	0.0	0:00.30	ksoftirqd/0
15	root	20	0	0	0	0	I	0.0	0.0	0:01.14	rcu_preempt
16	root	20	0	0	0	0	S	0.0	0.0	0:00.00	rcu_exp_par_gp_kthread_worker/0
17	root	20	0	0	0	0	S	0.0	0.0	0:00.10	rcu_exp_gp_kthread_worker
18	root	rt	0	0	0	0	S	0.0	0.0	0:00.18	migration/0
19	root	20	0	0	0	0	S	0.0	0.0	0:00.00	kprobe-optimizer
20	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/0
21	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/0
22	root	20	0	0	0	0	S	0.0	0.0	0:00.00	cpuhp/1
23	root	-51	0	0	0	0	S	0.0	0.0	0:00.00	idle_inject/1
24	root	rt	0	0	0	0	S	0.0	0.0	0:00.09	migration/1
25	root	20	0	0	0	0	S	0.0	0.0	0:02.23	ksoftirqd/1

Key Learning:

top helps monitor performance.

23. Create and manage user groups

Command Executed:

sudo groupadd developers

getent group developers

sudo usermod -aG developers ubuntu

cat /etc/group

Command Output:

Group created and user added.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ sudo groupadd developers
ubuntu@ip-172-31-13-215:~$ getent group developers
developers:x:1002:
ubuntu@ip-172-31-13-215:~$ sudo usermod -aG developers ubuntu
```

```
ubuntu:x:1000:
ssl-cert:x:109:
mysql:x:110:
tomcat:x:1001:
developers:x:1002:ubuntu
```

Key Learning:

Groups help manage permissions.

24. Set up SSH passwordless authentication**Command Executed:**

```
ssh-keygen -t ed25519
```

```
ssh-copy-id user@server
```

Command Output:

SSH key pair created and copied to remote server.

Evidence (Screenshot):

Screenshot of key generation and test login.

Key Learning:

SSH keys allow secure passwordless login.

25. Monitor log files using tail or grep**Command Executed:**

```
tail -f /var/log/syslog
```

```
grep "error" /var/log/syslog
```

Command Output:

Logs are shown live or filtered.

Evidence (Screenshot):

```
ubuntu@ubuntu2004:~/Desktop$ tail -f /var/log/syslog
2026-06-29T09:04:10.482001-04:00 ubuntu2004 gnome-shell[2081]: Invalid (NULL) pointer instance
2026-06-29T09:04:10.486000-04:00 ubuntu2004 gnome-shell[2081]: g_signal_connect_data: assertion 'G_TYPE_CHECK_INSTANCE (instance)' failed
2026-06-29T09:04:10.486071-04:00 ubuntu2004 gnome-shell[2081]: st_adjustment_get_values: assertion 'ST_IS_ADJUSTMENT (adjustment)' failed
2026-06-29T09:04:10.486949-04:00 ubuntu2004 gnome-shell[2081]: st_adjustment_get_values: assertion 'ST_IS_ADJUSTMENT (adjustment)' failed
2026-06-29T09:04:10.731674-04:00 ubuntu2004 gnome-shell[2081]: JS ERROR: TypeError: can't access property 'get_stage', this.label is null@012itemshowLabel@file:///usr/share/gnome-shell/extensions/ubuntu-dock@ubuntu.com/appicons.js:1374:23@012showLabel@file:///usr/share/gnome-shell/extensions/ubuntu-dock@ubuntu.com/dash.js:59:39@012syncLabel/this._showLabel@theoutid-gresource:///org/gnome/shell/ui/dash.js:546:38@012_init/this.timeout_add_once/id-gresource:///org/gnome/gjs/modules/core/overrides/GLib.js:436:13@012@resource:///org/gnome/shell/ui/init.js:120:20
2026-06-29T09:04:26.139803-04:00 ubuntu2004 systemd[2281]: pyxis-spam-3322000-e0e6-4f10-8b6f-a30f92082ae.scope: Consumed 7.149s CPU time over 55min 56.757s wall clock time, 54.3M memory peak.
2026-06-29T09:09:19.469817-04:00 ubuntu2004 kernel: workqueue: blk_mq_run_work_fn hogged CPU for +1333us 19 times, consider switching to WQ_UNBOUND
2026-06-29T09:10:04.879308-04:00 ubuntu2004 systemd[1]: Starting sysstat-collect.service - system activity accounting tool...
2026-06-29T09:10:05.839301-04:00 ubuntu2004 systemd[1]: sysstat-collect.service: Deactivated successfully.
2026-06-29T09:10:05.840305-04:00 ubuntu2004 systemd[1]: Finished sysstat-collect.service - system activity accounting tool.
```

Key Learning:

tail and grep are useful for troubleshooting.

26. Set up a web server

Command Executed:

```
sudo apt install apache2
```

```
sudo systemctl enable apache2
```

```
sudo systemctl start apache2
```

```
sudo systemctl status apache2
```

Command Output:

Apache is installed and running.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ sudo systemctl start apache2
ubuntu@ip-172-31-13-215:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
   Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-06-29 17:29:48 UTC; 1min 2s ago
 Invocation: 2b25512684ec47139a1a504a9fdcc7db
    Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2482 (apache2)
  Status: "Total requests: 0; Idle/Busy workers 100/0; Requests/sec: 0; Bytes served/sec: 0 B/sec"
    Tasks: 55 (limit: 627)
   Memory: 5.5M (peak: 5.9M)
      CPU: 88ms
   CGroup: /system.slice/apache2.service
           └─2482 /usr/sbin/apache2 -k start -DFOREGROUND
             └─2498 /usr/sbin/apache2 -k start -DFOREGROUND
               └─2499 /usr/sbin/apache2 -k start -DFOREGROUND
```

27. Configure and secure MySQL database

Command Executed:

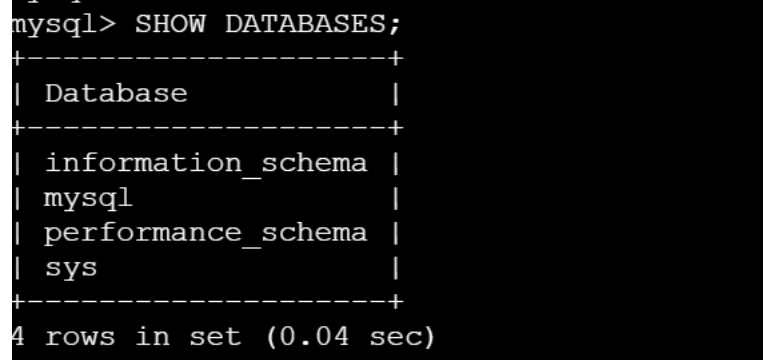
```
sudo apt install mysql-server
```

```
sudo mysql_secure_installation
```

```
mysql -u root -u
```

Command Output:

MySQL installed and secured.

Evidence (Screenshot):

```
mysql> SHOW DATABASES;
+-----+
| Database                |
+-----+
| information_schema      |
| mysql                   |
| performance_schema      |
| sys                     |
+-----+
4 rows in set (0.04 sec)
```

Key Learning:

mysql_secure_installation improves database security.

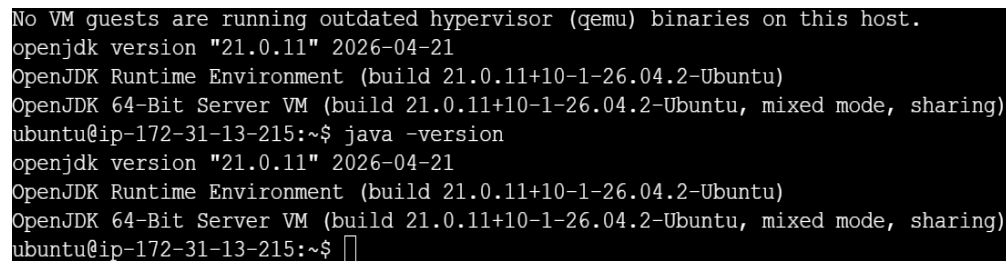
28. Set up Apache Tomcat application server

Command Executed:

```
sudo apt update
```

```
sudo apt install openjdk-21-jre-headless
```

```
java -version
```

Evidence (Screenshot):

```
No VM guests are running outdated hypervisor (qemu) binaries on this host.
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-13-215:~$ java -version
openjdk version "21.0.11" 2026-04-21
OpenJDK Runtime Environment (build 21.0.11+10-1-26.04.2-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.11+10-1-26.04.2-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-13-215:~$
```

29. Create a service file for Apache Tomcat

Command Executed:

```
sudo nano /etc/systemd/system/tomcat.service
```


Service File Content:

text

[Unit]

Description=Apache Tomcat Web Application Container

After=network.target

[Service]

Type=forking

User=tomcat

Group=tomcat

Environment=CATALINA_HOME=/opt/tomcat

Environment=CATALINA_BASE=/opt/tomcat

Environment=JAVA_HOME=/usr/lib/jvm/ openjdk-17-jre-headless

ExecStart=/opt/tomcat/bin/startup.sh

ExecStop=/opt/tomcat/bin/shutdown.sh

Restart=on-failure

[Install]

WantedBy=multi-user.target

Command Output:

Service file saved successfully.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:/tmp$ sudo nano /etc/systemd/system/tomcat.service
ubuntu@ip-172-31-13-215:/tmp$ sudo cat /etc/systemd/system/tomcat.service
[Unit]
Description=Apache Tomcat Web Application Container
After=network.target

[Service]
Type=forking
User=tomcat
Group=tomcat
Environment=CATALINA_HOME=/opt/tomcat
Environment=CATALINA_BASE=/opt/tomcat
Environment=JAVA_HOME=/usr/lib/jvm/java-17-openjdk-amd64
ExecStart=/opt/tomcat/bin/startup.sh
ExecStop=/opt/tomcat/bin/shutdown.sh
Restart=always

[Install]
WantedBy=multi-user.target
```

Key Learning:

systemd service files allow Tomcat to be managed with systemctl.

30. Start Tomcat using systemctl

Command Executed:

sudo systemctl daemon-reload

sudo systemctl enable tomcat

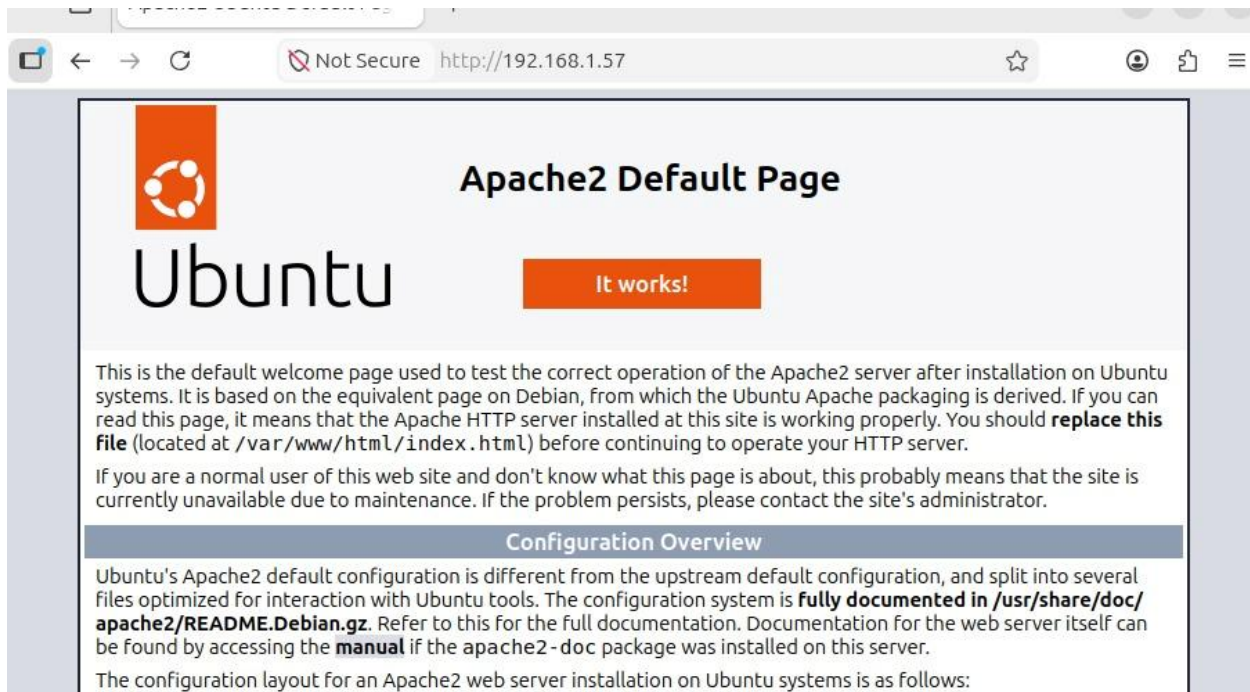
sudo systemctl start tomcat

sudo systemctl status tomcat

Command Output:

Tomcat service starts successfully.

Evidence (Screenshot):



31. Print specific columns from a delimited file

Command Executed:

```
awk -F, '{print $1, $7}' /etc/passwd
```

Command Output:

Selected columns are printed.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:/tmp$ awk -F: ' '{print $1, $7}' /etc/passwd
root /bin/bash
daemon /usr/sbin/nologin
bin /usr/sbin/nologin
sys /usr/sbin/nologin
sync /bin/sync
games /usr/sbin/nologin
man /usr/sbin/nologin
lp /usr/sbin/nologin
mail /usr/sbin/nologin
news /usr/sbin/nologin
uucp /usr/sbin/nologin
proxy /usr/sbin/nologin
www-data /usr/sbin/nologin
backup /usr/sbin/nologin
```

Key Learning:

AWK is useful for extracting columns.

32. Filter and print lines based on a pattern

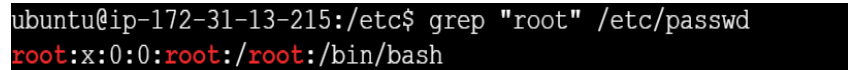
Command Executed:

```
grep "root" /etc/passwd
```

Command Output:

Only matching lines are printed.

Evidence (Screenshot):



```
ubuntu@ip-172-31-13-215:/etc$ grep "root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
```

Key Learning:

grep searches for patterns.

33. Calculate and print average, sum, or statistics

Command Executed:

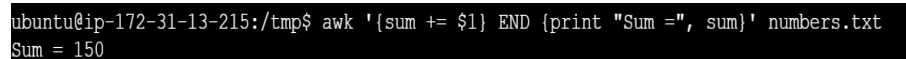
```
awk '{sum += $1} END {print "Sum =", sum}' numbers.txt
```

```
ls
```

Command Output:

Sum and average are displayed.

Evidence (Screenshot):



```
ubuntu@ip-172-31-13-215:/tmp$ awk '{sum += $1} END {print "Sum =", sum}' numbers.txt
Sum = 150
```

Key Learning:

AWK can perform arithmetic on columns.

34. Perform string manipulation

Command Executed:

```
echo "hello world" | tr '[:lower:]' '[:upper:]'
```

Command Output:

Text is printed in uppercase.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ echo "hello world" | tr '[:lower:]' '[:upper:]'
HELLO WORLD
```

35. Count the occurrences of a specific pattern

Command Executed:

```
grep -c "root" /etc/passwd
```

Command Output:

Number of matching lines is shown.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ grep -c "root" /etc/passwd
1
```

Key Learning:

grep -c counts matching lines.

36. Sort lines based on a specific field

Command Executed:

```
sort -t: -k3,3n /etc/passwd
```

Command Output:

Rows are sorted by the second field.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ sort -t: -k3,3n /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
```

Key Learning:

sort helps arrange data by columns.

37. Merge multiple files based on a common field

Command Executed:

```
cat > students.txt << EOF
```

```
101 Alice
```

```
102 Bob
```

```
103 Charlie
```

```
EOF
```

```
cat > marks.txt << EOF
```

```
101 85
```

```
102 90
```

```
103 78
```

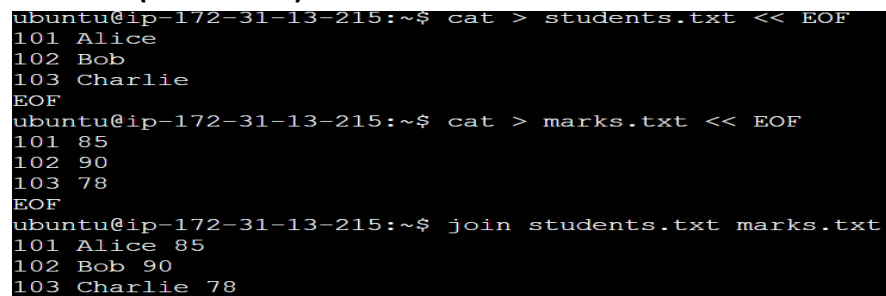
```
EOF
```

```
join students.txt marks.txt
```

Command Output:

Combined data is displayed.

Evidence (Screenshot):



```
ubuntu@ip-172-31-13-215:~$ cat > students.txt << EOF
101 Alice
102 Bob
103 Charlie
EOF
ubuntu@ip-172-31-13-215:~$ cat > marks.txt << EOF
101 85
102 90
103 78
EOF
ubuntu@ip-172-31-13-215:~$ join students.txt marks.txt
101 Alice 85
102 Bob 90
103 Charlie 78
```

Key Learning:

join combines files with common fields.

38. Substitute text in a file

Command Executed:

```
echo "Linux is good. Linux is fast." > sample.txt
```

```
sed -i 's/Linux/Ubuntu/g' sample.txt
```

```
cat sample.txt
```

Command Output:

Text is replaced in output.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ echo "Linux is good. Linux is fast." > sample.txt
ubuntu@ip-172-31-13-215:~$ sed -i 's/Linux/Ubuntu/g' sample.txt
ubuntu@ip-172-31-13-215:~$ cat sample.txt
Ubuntu is good. Ubuntu is fast.
```

Key Learning:

sed is used for search and replace.

39. Delete specific lines based on a pattern or line number**Command Executed:**

```
grep -v "root" /etc/passwd
```

```
sed '2,5d' /etc/passwd
```

Command Output:

Delete lines 2 through 5

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ sed '2,5d' /etc/passwd
root:x:0:0:root:/root:/bin/bash
games:x:5:60:games:/usr/games:/usr/sbin/nologin
man:x:6:12:man:/var/cache/man:/usr/sbin/nologin
lp:x:7:7:lp:/var/spool/lpd:/usr/sbin/nologin
mail:x:8:8:mail:/var/mail:/usr/sbin/nologin
news:x:9:9:news:/var/spool/news:/usr/sbin/nologin
uucp:x:10:10:uucp:/var/spool/uucp:/usr/sbin/nologin
```

Key Learning:

sed can delete lines by number.

40. Append or insert text before or after a pattern**Command Executed:**

```
cat > sample.txt << EOF
```

```
apple
```

```
banana
```

```
orange
```

EOF

sed '/banana/a\grapes' sample.txt

Command Output:

A line is inserted before the matched pattern.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ cat > sample.txt << EOF
apple
banana
orange
EOF
ubuntu@ip-172-31-13-215:~$ sed '/banana/a\grapes' sample.txt
apple
banana
grapes
orange
```

Key Learning:

sed can add text around a pattern.

41. Print only specific lines

Command Executed:

sed -n '2,5p' /etc/passwd

Command Output:

Only selected lines are printed.

Evidence (Screenshot):

```
ubuntu@ip-172-31-13-215:~$ sed -n '1,5p' /etc/passwd
root:x:0:0:root:/root:/bin/bash
daemon:x:1:1:daemon:/usr/sbin:/usr/sbin/nologin
bin:x:2:2:bin:/bin:/usr/sbin/nologin
sys:x:3:3:sys:/dev:/usr/sbin/nologin
sync:x:4:65534:sync:/bin:/bin/sync
```

Key Learning:

sed -n prints only requested lines.

42. Copy file from Linux to Windows machine

Command Executed:

bash

scp file.txt windowsuser@windows-ip:/path/

Command Output:

File transferred successfully.

Evidence (Screenshot):

Screenshot of transfer command.

Key Learning:

scp or pscp is used to transfer files between Linux and Windows.[stackoverflow+1](#)
